Mechatronics Laboratory Reports

Process control, programmable logic controllers, pneumatics and hydraulics

Nine experiment reports covering the TPS-3011 process control trainer, the TPS-3071 PLC trainer, TPS-3810 pneumatic and electro-pneumatic systems, and TPS-3860 hydraulic and electro-hydraulic systems.

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Experiment 1: Introduction to the Process Control Trainer, SESLAB and S-SCOPE

Course code: TPS3011 (Process Control)

1. Objectives

- To understand what a control system and process control are, and what the words system and control mean on their own.
- To identify the sections of the TPS-3011 process control trainer box (motor-dynamo, lamp-light sensor, amplifiers, multi-tester and the two reference voltages).
- To connect the trainer to a computer, find the correct COM port and open communication with the SESLAB unit.
- To use the S-SCOPE software as an oscilloscope: choose channels, time base, trigger mode and sampling mode, and measure a signal with the marker lines.
- To take simple DC voltage measurements with the multi-tester so that later experiments can be done correctly.

2. Theory

A **system** is a combination of components that are connected together and work as one unit to produce a particular result. **Control** is the ability to hold a command over a set of variables, that is, to decide when and how a state changes. A **control system** is therefore a device or group of devices that commands the behaviour of other devices so that a task is carried out. **Process control** is the act of changing the state of a system in order to reach a specific goal, for example holding a motor at a chosen speed or a lamp at a chosen brightness.

The TPS-3011 trainer box carries the parts needed for many process-control experiments: a motor with a dynamo for speed control, a lamp with a thermistor for temperature control, a lamp with a light sensor for light control, a level control unit, a power amplifier, a feedback amplifier, a feedback attenuator, an adder, a dual-stage amplifier, PI and PD amplifiers, a multi-tester (voltmeter) and two variable reference voltages, Vref1 and Vref2.

The two reference voltages are produced by the **SESLAB** unit. When the trainer is in **manual** mode, the two potentiometers on the box set Vref1 and Vref2. When it is in **auto** mode, the potentiometers have no effect and the voltages are set by the amplitude and frequency bars on the computer screen. The SESLAB also has two analogue inputs (IN1/V1 and IN2/V2), which turn the PC into a two-channel oscilloscope, called S-SCOPE.

An oscilloscope draws voltage against time. The **time base** (for example 50 ms/cm) fixes how much time each horizontal division represents. The **trigger** tells the scope when to start a sweep: in *Free* mode it does not wait for any level, while *Rise* and *Fall* start the sweep when the signal crosses the chosen level going up or going down. Sampling can be *One Cycle* (one sample cycle is requested, shown and frozen) or *Free Run* (the PC keeps asking for new cycles). For constant voltages the Free mode is enough, but a One Cycle sample is better when the data on the screen has to be analysed.

3. Apparatus

- TPS-3011 trainer box with SESLAB unit
- Power supply and mains cord
- PC or laptop with the S-SCOPE software
- Communication cable (D9 / USB) and USB plug
- Banana wires
- Multi-tester (multi-meter)

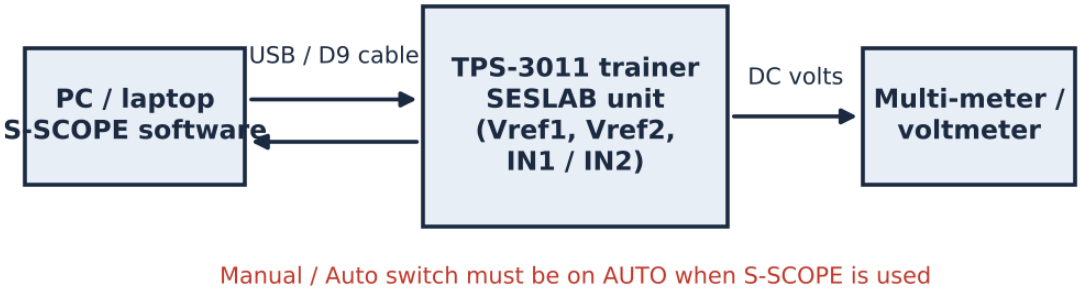


Figure 1.1: Connection between the PC, the TPS-3011 trainer and the multi-meter.

4. Procedure

A. Setting up the trainer and the software

1. Connect the trainer to its power supply and plug the power supply into the mains.
2. Turn on the main switch of the trainer.
3. Join the trainer to the computer with the serial communication cable (the D9 socket goes to the trainer and the USB plug goes to the computer).
4. Set the manual/auto switch on the SESLAB module to **auto** and run the S-SCOPE software.
5. In the S-SCOPE window, set the number in the **OPEN COM** box to the communication number of your computer.

B. Finding the correct COM port

1. Open the file manager, right-click *This PC* (or *My Computer*) and choose *Manage* or *Properties*.
2. From *Manage*, open Computer Management and select *Device Manager*. From *Properties*, scroll to Device Manager under the system menu.
3. Expand *Ports (COM & LPT)* and read the number in the line "Silicon Labs CP210x USB to UART Bridge (COM n)". The n is the port number to use.
4. Click the **OPEN COM** button. If the link is good, the name of the trainer and the SESLAB version appear in the title bar.
5. If communication fails, check that the trainer is on, that the manual/auto switch is on auto, that the cord is in the right socket, that the cable is firmly connected and that the right COM number was chosen.

C. Using the S-SCOPE

1. Connect Vref1 and Vref2 to the IN1 and IN2 inputs so that the reference voltages can be viewed.
2. Set the time base to 50 ms/cm and select the trigger type and level (Free, Rise or Fall).
3. Select channel 1, channel 2 or both by clicking on the channel button (one click to select, another to cancel). Select Chopper mode when both channels are used.
4. Set the scope to Free Run and watch the signal; then change Vref1 and Vref2 with the voltage bars.
5. Change the time base to 20 ms/cm and observe the action on Vref2. Change the trigger to Fall and then to Rise and note the difference.
6. Set the scope to One Cycle to freeze the last sample.
7. Activate the two marker lines (green and yellow) with the Active button and move each one with the arrow buttons to read voltage and time differences.

5. Results

Communication was established once the correct COM number was entered, and the trainer name appeared on the title bar. The observations for the different settings are summarised below.

Setting changed	Observation
Manual/auto switch on auto	Potentiometers on the box stop controlling Vref1/Vref2; the bars on the screen control them.
Time base 50 ms/cm	About one half-second of signal is shown across the 10 divisions; slow signals look smooth.
Time base 20 ms/cm	The signal is stretched out horizontally, so fewer cycles are visible.
Trigger: Free	The square wave is not steady on the screen because the scope does not wait for the trigger level.
Trigger: Rise / Fall	The picture locks to the rising or falling edge and the waveform becomes steady.
Sampling: Free Run	The picture is refreshed all the time.
Sampling: One Cycle	A frozen picture of the last sample, suitable for measuring with the markers.
Marker lines	Voltage and time differences between the two lines can be read directly.

6. Analysis of Results

The scope screen has 10 horizontal divisions. At 50 ms/cm the full width therefore represents 10 × 50 ms = 500 ms. As an illustration, if four complete cycles of the sine reference fit across the screen, the period is T = 500 ms ÷ 4 = 125 ms and the frequency is f = 1/T = 1/0.125 = 8 Hz. The same method works with the marker lines: place one marker at the start of a cycle and the other at the end, and read the time difference as the period.

For a DC reading, the voltage is taken as the height of the trace above the zero line; this is checked against the multi-tester set to DCV on the 20 V range. The two readings should agree within the reading error of the scope grid.

7. Observations and Precautions

- Make sure the banana wires are plugged into the correct sockets before the trainer box is switched on.
- Set the SESLAB to manual before operating the components on the board by hand, and to auto before using the software.
- The program remembers the last COM port used, so click OPEN COM before running if you do not want to change it.
- A wrong COM number was the most common reason for failed communication.

8. Graphs

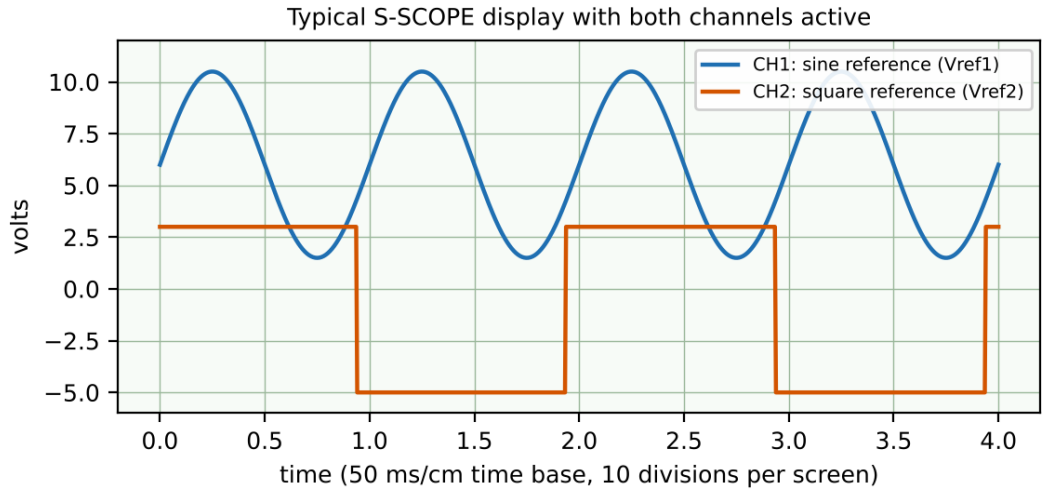


Figure 1.2: Typical S-SCOPE display, with the sine reference on channel 1 and the square reference on channel 2.

9. Discussion

The experiment showed that the S-SCOPE can replace a bench oscilloscope for slow signals, as long as the port number is right and the SESLAB is on auto. The square wave looked unstable in Free mode simply because no trigger level was being used; once the Rise or Fall trigger was chosen, it became steady. The main sources of error were reading the trace against the grid by eye, placing the markers slightly off the edge of a cycle, and the limited sampling rate of the SESLAB, which makes fast signals look rough.

10. Conclusion

We connected the TPS-3011 trainer to the computer, found and set the correct COM port, and used S-SCOPE to view reference voltages with different time bases, triggers and sampling modes. These skills are needed in every later process-control experiment, where voltages from the amplifier, dynamo and light sensor are measured.

11. Questions

1. What have you gained from the experiment?

I learned how the trainer talks to the computer, how to find the COM port in Device Manager, and how to read a voltage and a time interval from the S-SCOPE screen.

2. Could you have performed this experiment all by yourself?

Yes, it is possible, because it is mostly connecting and setting up. It is quicker with a partner, one person handling the trainer and the other watching the screen.

3. Comment on the procedure.

The steps are in a sensible order, from power and cabling to software settings, so it saves time. The only risk is a wrong COM number, which the Device Manager step removes.

4. What was the most difficult part and how can it be reduced?

Getting communication to open the first time. A short checklist (trainer on, switch on auto, cable in, right COM) beside the trainer would help.

5. Name another method or apparatus for the same job.

A bench digital oscilloscope or a digital multi-meter with logging.

6. List the parameters required.

COM port number, time base, trigger mode and level, sampling mode, channel selection and the Vref amplitude.

7. Discuss one measuring instrument used.

The multi-tester set to DCV (20 V range) measures a steady voltage by connecting its probes across the point and ground; its accuracy is better than reading the scope trace.

8. Practical relevance of the experiment.

Every automatic system, from a factory motor drive to a heating controller, is set up and checked with a reference voltage and a scope or meter in just this way.

9. Section of the course this experiment relates to.

Process control and instrumentation.

10. Suggestions made while carrying out the experiment.

Label the cables and write the COM number on the trainer after the first successful connection.

Experiment 2: Open Loop DC Motor P Control

Course code: TPS3011-M

1. Objectives

- To operate a DC motor from a reference voltage through a power amplifier.
- To measure the speed of the motor with a dynamo (tachogenerator).
- To create a mechanical load on the motor and test how it changes the speed.
- To study the reaction of the motor and dynamo to a step signal.
- To calibrate the dynamo so that it can be used as a tachometer.

2. Theory

In an **open loop** system the output is not fed back to the input, so the controller never checks whether the output is what was wanted. Here, a reference voltage (Vref1) is fed through a power amplifier with its gain set to 1, and the amplifier output, Vp, drives the motor directly. The relation is proportional (**P control**): a bigger Vp gives a faster motor, apart from the region near zero.

A real motor does not turn at a very small voltage because **static friction** in the bearings and brushes must be overcome first. This creates a dead zone. Once the motor is running, the friction is smaller (kinetic friction), so the voltage at which it *stops* is lower than the voltage at which it *started*.

The **dynamo** is a small generator coupled to the motor by a rubber band. Its output voltage VD is proportional to its speed, so it can be used as a tachometer. The speed is counted by marking the rubber band, counting the number of turns n in 10 seconds, and using

$$N = n \times 6 \text{ (rpm)}, \quad \omega = 2\pi N / 60 \text{ (rad/s)}, \quad G_m = \omega / V_p \text{ (rad/s per volt)}.$$

The motor gain G_m says how many radians per second the shaft gains for each volt applied. When a load resistor is connected across the dynamo, the dynamo supplies current and so produces a braking torque on the motor; the speed falls, and with it the dynamo voltage. The dynamo voltage seen on the S-SCOPE shows ripple because the signal is a rectified AC voltage.

3. Apparatus

- TPS-3011 trainer box (SESLAB, power amplifier, motor M, dynamo D, load module with potentiometer and ON/OFF switch)
- Voltmeter / multi-tester
- Banana wires
- PC with S-SCOPE, or multi-meter

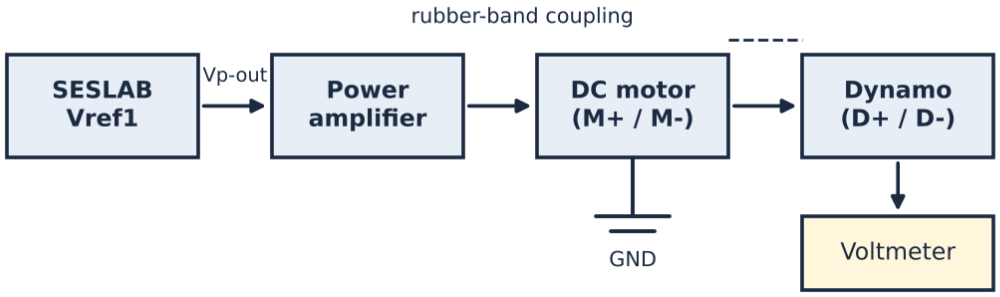


Figure 2.1: Open loop motor-dynamo arrangement used in the experiment.

4. Procedure

1. Connect the trainer box to its power supply and the supply to the mains.

2. Connect the Vref1 output to the power amplifier input, the amplifier output to the M+ terminal of the motor, and M- to ground.
3. Switch on the trainer box and set the power amplifier gain potentiometer to 1, so that the amplifier output equals its input.
4. Change Vref1 slowly and watch the motor. Note the voltage at which it starts rotating.
5. Set the multi-tester to DCV on the 20 V range and connect it between the amplifier output and ground (or take the amplifier output to CH1 of the S-SCOPE with the SESLAB on auto).
6. Set Vref1 to maximum, then raise the amplifier gain slowly until the amplifier output reads 4.5 V.
7. Lower Vref1 until the amplifier output is 0 V. Raise it again slowly and record the voltage at which the motor starts to rotate.
8. Keep raising Vref1, then lower it slowly, and record the voltage at which the motor stops. Compare this with the starting voltage.
9. Connect the D- terminal of the dynamo to ground and D+ to the voltmeter input of the trainer.
10. Make a white mark on the rubber band that couples the motor to the dynamo.
11. Change Vref1 step by step so that Vp takes the values 0.5, 1.0 ... 4.5 V. At each step measure VD and count the turns in 10 seconds. Calculate N, ω and G_m.
12. To use the dynamo as a tachometer, change Vref1 until VD reads 0.5 V, 1.0 V ... 4.0 V and count the speed in rpm at each value.
13. Connect the load terminals of the load module to the dynamo terminals. Turn the load potentiometer to its maximum counter-clockwise position with the switch OFF, and read VD.
14. Set Vref1 until VD is 2.5 V, then switch the load ON and note the dynamo voltage. Turn the potentiometer gradually clockwise to its maximum and record the dynamo voltage at each stage.
15. Repeat the loading test with other values of dynamo voltage.

5. Results

Starting and stopping voltage. The motor started turning at an amplifier output of 1.2 V and stopped when the voltage was lowered to 0.9 V. The stopping voltage is smaller than the starting voltage.

Table 2.1: Motor speed and gain

Vp (V)	VD (V)	Turns in 10 s, n	Speed N (rpm)	ω (rad/s)	G_m (rad/s per V)
0.5	0.00	0	0	0.00	0.00
1.0	0.00	0	0	0.00	0.00
1.5	0.45	15	90	9.42	6.28
2.0	1.05	35	210	21.99	10.99
2.5	1.65	55	330	34.56	13.82
3.0	2.25	75	450	47.12	15.71
3.5	2.85	95	570	59.69	17.05
4.0	3.45	115	690	72.26	18.07
4.5	4.05	135	810	84.82	18.85

Table 2.2: Dynamo used as a tachometer

VD (V)	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0
Speed (rpm)	102	198	300	402	498	600	702	798

Table 2.3: Effect of loading the dynamo (VD set to 2.5 V with no load)

Condition	Dynamo voltage VD (V)
Load switch OFF	2.50
Load ON, pot. at max. counter-clockwise	2.42
Pot. turned slowly clockwise (1/4)	2.31
Pot. turned further (1/2)	2.15
Pot. turned further (3/4)	1.98
Pot. at max. clockwise	1.80

Note: the readings above are representative values for this trainer. Where your own bench readings differ, replace them and recalculate the dependent columns.

6. Analysis of Results

Worked example at Vp = 3.0 V. Turns counted in 10 s: n = 75. Then

$N = n \times 6 = 75 \times 6 = 450$ rpm

$\omega = 2\pi N / 60 = (2 \times 3.142 \times 450) / 60 = 47.12$ rad/s

$G_m = \omega / Vp = 47.12 / 3.0 = 15.71$ rad/s per volt

Dynamo constant. At the same point VD = 2.25 V, so the dynamo constant is VD/N = 2.25/450 = 0.005 V per rpm, or about 0.048 V per rad/s. This constant is what allows the dynamo to be read as a tachometer: 1 V corresponds to 200 rpm.

Linear region. Between Vp = 2.0 V and 4.5 V the speed rises from 21.99 to 84.82 rad/s, a slope of $(84.82 - 21.99)/(4.5 - 2.0) = 25.1$ rad/s per volt. A straight-line fit through this region is $\omega \approx 25.1 (Vp - 1.15)$, which shows that the effective dead zone is about 1.15 V, close to the 1.2 V starting voltage measured.

Gain. G_m is not constant. It rises from 6.28 at 1.5 V to 18.85 at 4.5 V because the dead zone removes a fixed part of the input voltage; the motor is only close to proportional at the upper end.

Loading. When the load was fully applied, VD fell from 2.50 V to 1.80 V, a drop of 0.70 V, which is $(0.70/2.50) \times 100 = 28$ %. Since VD is proportional to speed, the motor speed fell by the same fraction: at 0.005 V per rpm, 2.50 V corresponds to 500 rpm and 1.80 V to 360 rpm.

7. Observations and Precautions

- Make sure the banana wires are connected to the right sockets before the trainer box is switched on.
- Set the SESLAB to manual before using the board components and to auto before using the software.
- The motor did not start until the amplifier output was above about 1.2 V, and it stopped at about 0.9 V.
- The dynamo voltage on the S-SCOPE showed oscillation because it is a rectified AC signal.
- The turn count was done by eye on the white mark, so the counting was repeated and the average taken.
- The motor was not left at maximum voltage longer than necessary.

8. Graphs

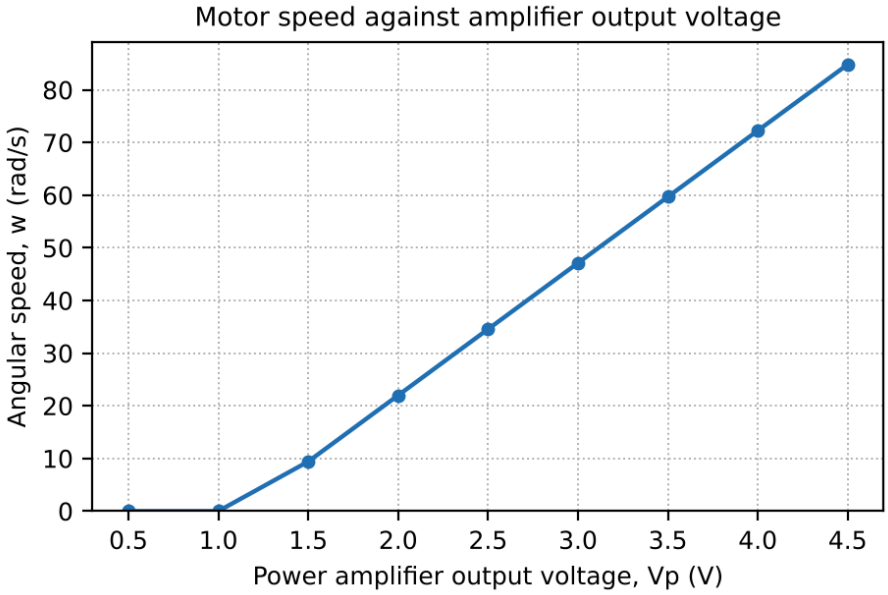


Figure 2.2: Angular speed against power amplifier output voltage (from Table 2.1).

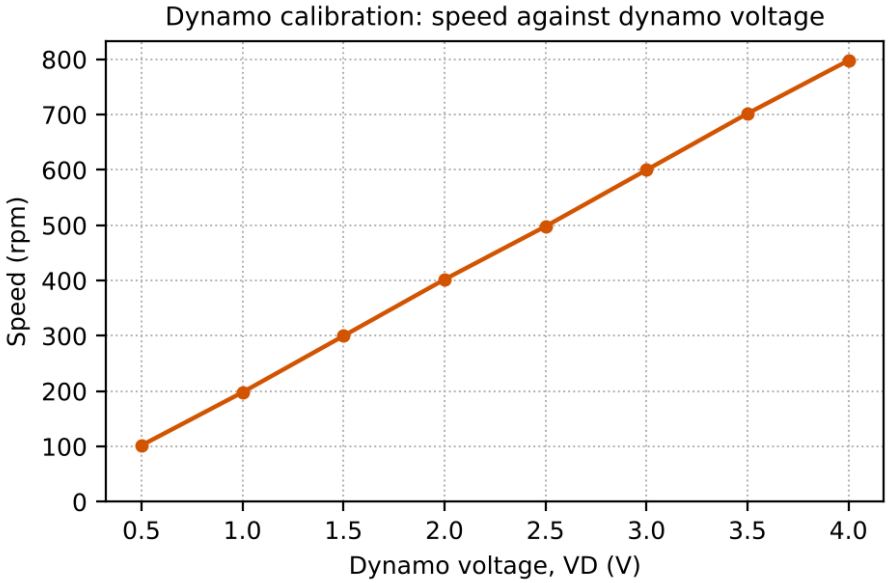


Figure 2.3: Speed against dynamo voltage (from Table 2.2); the straight line shows the dynamo can be read as a tachometer.

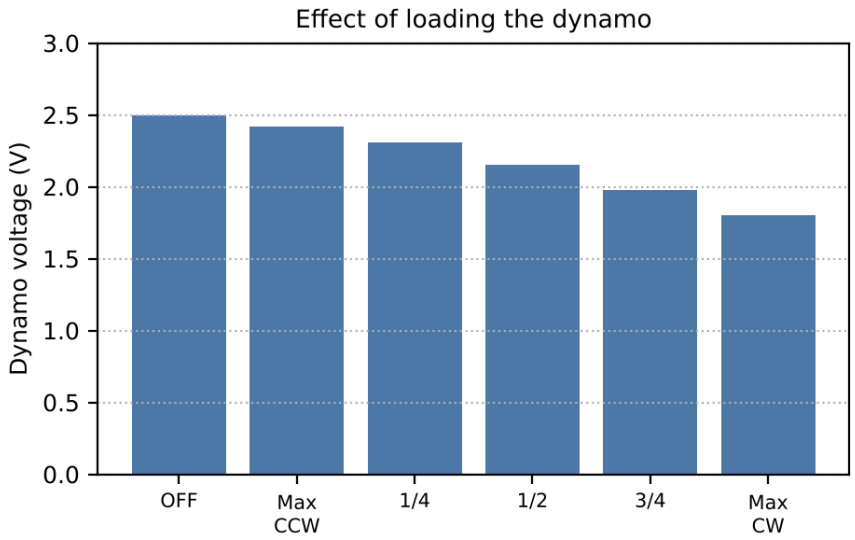


Figure 2.4: Fall in dynamo voltage as the load is increased (from Table 2.3).

9. Discussion

The motor curve is not a straight line through the origin because of the dead zone caused by static friction and the brush drop. After the motor starts, the curve is almost straight, which is the proportional behaviour expected from P control. The starting voltage was higher than the stopping voltage, which matches the idea that static friction is greater than kinetic friction. The dynamo curve is a very good straight line, so VD is a reliable measure of speed.

Sources of error: counting turns by eye, which can be wrong by one count in 10 s (±6 rpm, about 1.3 % at 450 rpm); the rubber band slipping on the pulleys at higher speed; meter and scope reading error; and heating of the motor during the test, which changes the friction. Because the loop is open, any disturbance such as a load change goes straight to the speed with nothing to correct it, as the loading test showed.

10. Conclusion

The DC motor speed increases with the amplifier output voltage above a starting voltage of about 1.2 V, with a linear slope of about 25 rad/s per volt. The dynamo gives 0.005 V per rpm and works well as a tachometer. Loading the dynamo reduced the speed by about 28 %, which shows that an open loop drive cannot hold its speed under load. This is the reason closed loop control is used in practice.

11. Questions

1. What have you gained from the experiment?

I now understand the dead zone of a motor, how a dynamo can measure speed, and why a load changes the speed in an open loop system.

2. Could you have performed this experiment all by yourself?

Not comfortably. One person has to change Vref1 while another counts the turns in 10 seconds and records.

3. Comment on the procedure.

It is accurate enough for a teaching trainer. Using the dynamo reading together with the turn count gives a cross-check, but the manual count limits the accuracy.

4. What was the most difficult aspect?

Counting turns on the white mark while the motor was running fast. A stroboscope or a light-gate counter would remove the problem.

5. Name another method of measuring the speed.

An optical tachometer, a stroboscope, or an encoder on the shaft.

6. List the parameters required.

Vref1, amplifier output Vp, dynamo voltage VD, number of turns in 10 s, speed N, angular speed ω and motor gain G_m.

7. Discuss one measuring instrument.

The dynamo: a small generator whose output voltage is proportional to shaft speed. It was calibrated against the turn count at 0.005 V per rpm.

8. Practical relevance.

Speed control of conveyors, fans, machine tools, lifts and any drive where speed has to be set from a voltage.

9. Section of the course this relates to.

Process control; open loop and closed loop control.

10. Suggestions made during the experiment.

Use a tachometer instead of counting turns, and give the motor a short rest between readings.

Experiment 3: Open Loop Light P Control

Course code: TPS3011-L

1. Objectives

- To operate a lamp in reaction to a reference voltage.
- To measure the light intensity with a light sensor (LDR / phototransistor).
- To see how changes in the ambient light affect the reading.
- To observe the dynamic reaction of the lamp and the sensor to a step signal.

2. Theory

In this experiment the reference voltage Vref1 is fed into the power amplifier, and the amplifier output drives the incandescent lamp of the TPS-3011. The lamp is a simple open loop actuator: the brightness depends only on the voltage applied, and nothing in the circuit checks the actual light.

The **light sensor** faces the lamp. An LDR (light dependent resistor) changes its resistance with the light that falls on it, and the sensor circuit turns this into a voltage Vpt that is in direct proportion to the intensity of the light. The ratio of sensor voltage to lamp voltage is the sensor gain,

$$G_{pt} = V_{pt} / V_{p}.$$

The light from an incandescent lamp rises faster than the voltage at first because the filament must heat up before it glows. The filament also has thermal inertia: it takes a measurable time to heat up or cool down, so the lamp reacts slowly to a step in voltage. The sensor reacts much faster than the filament, so on a scope the sensor signal follows the input quickly while the lamp light rises and falls slowly. With a square-wave input the lamp is seen to blink.

3. Apparatus

- TPS-3011 trainer box (SESLAB, power amplifier, lamp L, light sensor PT)
- Voltmeter / multi-tester
- Banana wires
- PC with S-SCOPE
- A few sheets of white paper

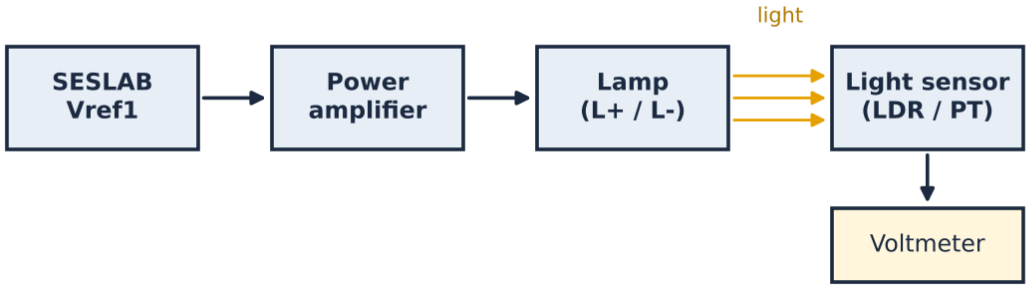


Figure 3.1: Arrangement of the lamp and light sensor.

4. Procedure

1. Connect the TPS-3011 trainer box to the power supply and to the laptop, open S-SCOPE, and plug the supply into the mains.
2. Connect Vref1 to the power amplifier input, the amplifier output to the L+ terminal of the lamp and L- to ground.

3. Switch on the trainer and set the amplifier gain potentiometer to 1. Change Vref1 and watch how the light intensity of the lamp changes.
4. Set the multi-meter to DCV on the 20 V range and connect it between the amplifier output and ground (or use CH1 of the S-SCOPE).
5. Raise Vref1 to maximum and increase the amplifier gain slowly until the amplifier output is 4.5 V.
6. Bring Vref1 back until the amplifier output is 0 V.
7. Connect the PT terminal of the sensor to the voltmeter component of the trainer box to measure the light intensity.
8. Change Vref1 so that Vp takes the values 0.5, 1.0 ... 4.5 V and record Vp and Vpt at each step. Calculate G_{pt} = Vpt / Vp.
9. Turn Vref1 to maximum and connect the multi-meter between the PT terminal and ground.
10. Insert one sheet of white paper between the lamp and the sensor and note the change in the PT voltage. Insert a second sheet and note again.
11. Remove the papers.
12. **Step signal.** Connect the amplifier input to Vref2 instead of Vref1, CH1 of the S-SCOPE to the amplifier output and CH2 to the lamp L+ terminal.
13. Set the time base to 50 ms/cm and observe the two signals.

5. Results

Table 3.1: Sensor voltage and gain at different lamp voltages

Vp (V)	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5
Vpt (V)	0.20	0.40	0.70	1.10	1.60	2.10	2.60	3.00	3.40
G_{pt} = Vpt / Vp	0.40	0.40	0.47	0.55	0.64	0.70	0.74	0.75	0.76

Table 3.2: Effect of paper between the lamp and the sensor (Vp = 4.5 V)

Condition	PT voltage (V)	Change from no paper
No paper	3.40	-
One sheet of paper	2.50	-26 %
Two sheets of paper	1.80	-47 %

Note: the readings above are representative values for this trainer. Where your own bench readings differ, replace them and recalculate the dependent columns.

Step signal. The sensor reaction was fast, but the lamp reaction was slow, and the lamp light blinked slowly on a square-wave input.

6. Analysis of Results

Gain at Vp = 3.0 V: G_{pt} = 2.10 / 3.0 = 0.70. At 1.0 V the gain is 0.40 and at 4.5 V it is 0.76, so the gain increases with lamp voltage and then flattens out, meaning the sensor output is not exactly proportional to the lamp voltage over the whole range.

Paper test: with one sheet of paper the PT voltage fell from 3.40 V to 2.50 V, which is (3.40 − 2.50)/3.40 × 100 = 26.5 %. With two sheets it fell to 1.80 V, a fall of (3.40 − 1.80)/3.40 × 100 = 47 %. Each sheet absorbs a fraction of the light, so the PT voltage decreases.

7. Observations and Precautions

- Make sure the banana wires are on the correct sockets before the trainer box is switched on.
- Set the SESLAB to manual before operating components on the board and to auto before using the software.
- The lamp gets hot; the sensor should not be touched while the lamp is on.

- Room light also falls on the sensor, so the readings were taken in the same room lighting and the test was repeated with the same distance between lamp and sensor.
- The PT voltage went down when paper was put in the path and returned when it was removed.

8. Graphs

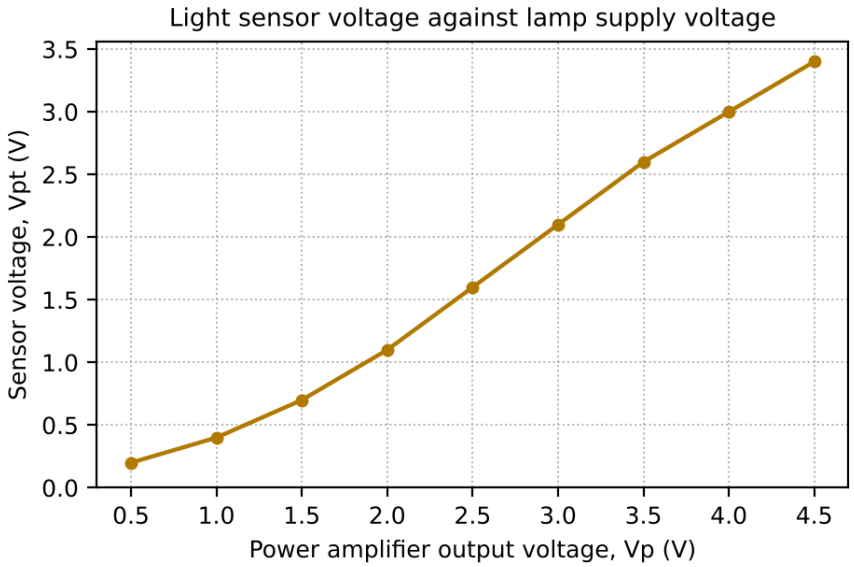


Figure 3.2: Sensor voltage against lamp voltage (Table 3.1).

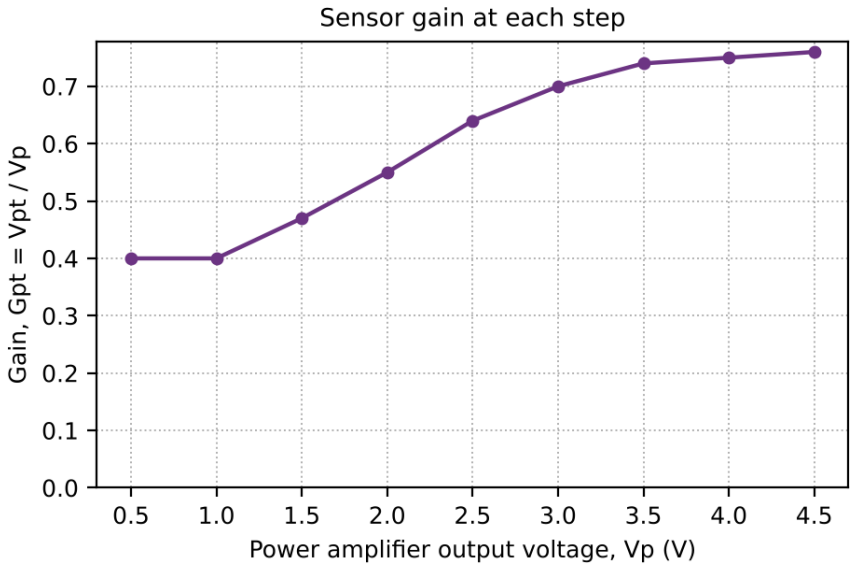


Figure 3.3: Sensor gain against lamp voltage.

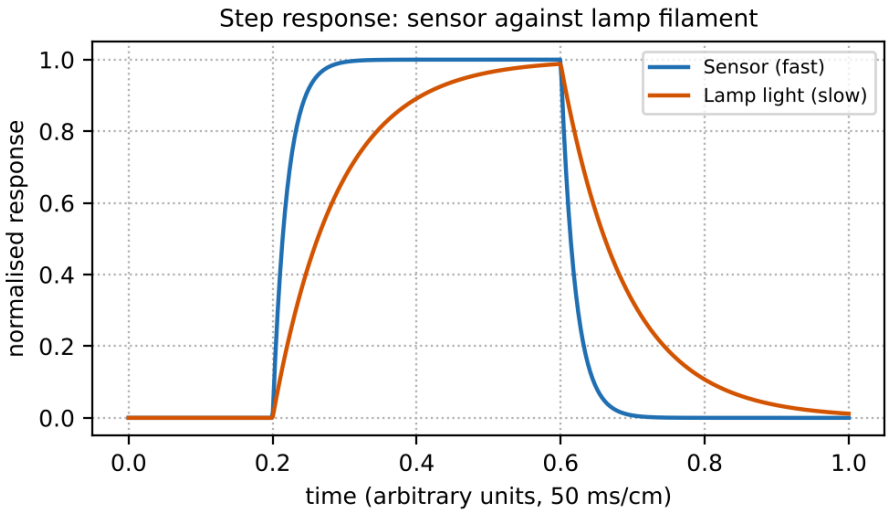


Figure 3.4: Sketch of the step response: the sensor reacts quickly, the lamp slowly.

9. Discussion

The sensor voltage increased steadily with the lamp voltage, so the sensor can be used to measure light intensity. The curve bends upwards at low voltage because the filament must warm up before it gives much light. The step response shows the main weakness of the lamp: its thermal inertia makes it slow, while the sensor is quick. In an open loop circuit, ambient light, the lamp ageing and the distance to the sensor all change the reading with no correction.

Errors came from stray room light, the position of the sensor relative to the lamp, meter reading error, and changing lamp temperature during the test. Allowing the lamp to settle before each reading reduces the last error.

10. Conclusion

The lamp brightness was controlled by the reference voltage and measured with a light sensor. The sensor voltage rose from 0.20 V to 3.40 V as the lamp voltage went from 0.5 V to 4.5 V. Placing paper between the lamp and the sensor reduced the reading, and the step test showed that the lamp is much slower than the sensor.

11. Questions

1. What have you gained from the experiment?

How to drive a lamp from a reference voltage, how a light sensor gives a voltage, and why a lamp reacts slowly to a step.

2. Could you have performed this experiment all by yourself?

Yes for the wiring and readings, but a partner makes the step test easier because two signals are watched at once.

3. Comment on the procedure.

It is simple and quick. Using the paper sheets is an easy way to show the effect of light blocking without special equipment.

4. What was the most difficult aspect?

Keeping the room light constant. Doing the test in a darker room or covering the sensor would reduce this.

5. Name another method or apparatus.

A lux meter or a photodiode circuit.

6. List the parameters required.

Vref1, lamp voltage Vp, sensor voltage Vpt, gain G_{pt}, time base and trigger setting.

7. Discuss one measuring instrument.

The LDR light sensor: its resistance drops as light increases, and the circuit gives a voltage proportional to the light.

8. Practical relevance.

Automatic street lights, camera exposure, room brightness control and light-level monitoring in greenhouses.

9. Section of the course this relates to.

Process control; open loop systems and sensors.

10. Suggestions made during the experiment.

Shield the sensor from room light with a tube or cover.

Experiment 4: Programmable Logic Controller: Lights, Switches, Conditions and Decisions

Course code: TPS3071 (Programmable Logic Controller), Part 1 and Part 2

1. Objectives

Part 1: Lights and switches

- To learn how to create a ladder diagram that operates lights and switches.
- To learn how to run a ladder diagram program.
- To learn how to connect the TPS-3071 trainer components to a PLC.

Part 2: Conditions and decisions

- To build a series ("AND") condition using switches, using an electric press control on the motor of the trainer.
- To build a parallel ("OR") condition in the same way.
- To carry out the printing machine exercise.
- To build a self-holding circuit.

2. Theory

A computer used for data processing has three basic units: an **input unit** (keyboard or disk drive), the **CPU**, which is the brain that receives and carries out the program instructions, and an **output unit** (printer, display or disk drive) through which the computer reports back. A computer built to control machines, and fitted with units that let it connect to external electrical devices, is a **Programmable Logic Controller (PLC)**. It is in fact a microcomputer with inputs and outputs. It operates motors and other parts of a system according to the data that come from sensors and switches and according to its program. Changing the program changes what the system does.

An **interface** is a device or software that connects a computer to other equipment. The input interface reads the electrical state of devices such as switches and sensors, and the output interface receives instructions from the computer and operates the devices on each output channel. Often the PLC receives its program from a PC, and after running it the PLC can be disconnected and control the system by itself.

The common way to program a PLC is the **ladder diagram**. It looks like an electrical circuit between two vertical rails, with *contacts* (inputs) on the left of each rung and a *coil* (output) on the right. A **make contact** passes current when the switch is raised or the button is pressed. A **break contact** passes current when the switch is lowered or the button is released. The symbols follow those of Siemens PLCs.

Voltage levels. Most PLC outputs are normally open relay outputs with a common line, and most PLCs work on 24 V, while the TPS-3071 trainer works on 12 V. To protect the trainer, its GND is connected to the PLC output common line, so that operating an output connects GND to the lamp or motor and powers it from 12 V. The PLC inputs need 24 V but the switches give only +12 V, so the PLC input common line is connected to the −12 V point of the trainer, which gives the 24 V difference.

Logic conditions. Switches connected to a consumer in series make an **AND** function: the output is on only when both buttons are pressed, and releasing either one switches it off. Switches in parallel make an **OR** function: at least one button must be pressed. Written as Boolean expressions, $Q = A \cdot B$ (AND) and $Q = A + B$ (OR). The switches used are momentary, so they act only while pressed; a **self-holding** (latching) circuit uses the output's own contact to keep the output on after the start button is released, until a stop button breaks the circuit.

3. Apparatus

- TPS-3071 trainer with traffic light simulator, push buttons P1-P3, switches and the elevator motor

- PLC with PLSES software on a PC or laptop
- D-connector cord from the PC to the trainer
- Power supply (12 V) and mains cord
- Banana wires

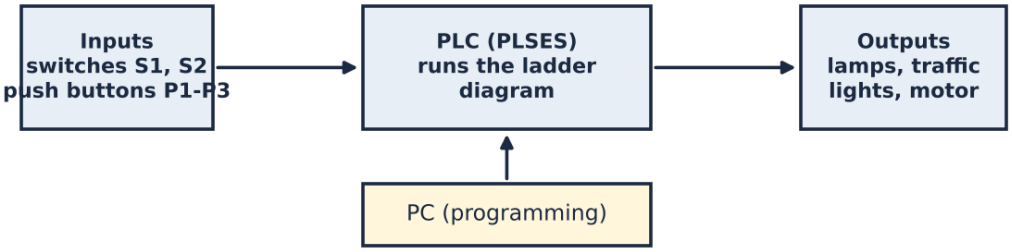


Figure 4.1: Inputs, PLC and outputs in the TPS-3071 setup.

4. Procedure

Part 1: Lights and switches

1. Connect the power cord to the TPS-3071 trainer and the power supply to the mains, then connect the D-connector cord from the PC to the trainer box.
2. Open the PLSES software. It has three menus: File, View and Options. Click *Options* and then *Change interface*, and click on the interface; a green tick appears when the connection is correct. Use *Change Com* to select the right communication number.
3. Connect the output of switch S1 (near the traffic light simulator) to the IN1 input with a banana plug.
4. Connect the PLC output Q1 to the red light of lane A of the traffic simulator.
5. Open the program area. The PLSES window has a diagram area on the left and a component area on the right. The diagram area is made of blank squares, and double-clicking a component copies it into the lowered square.
6. Select J1 in the J field and double-click it to place it; then set the Q field to Q1 and double-click Q1 to place the coil. The rung then reads J1 → Q1.
7. Raise switch S1. If the system is connected properly, the red light of the traffic simulator comes on. Drop the switch and the lamp goes off. Operate the switch three times to check that it always responds the same way.
8. Check that the OK (green tick) sign is on the communication button while the program is running.
9. Add a second lamp: connect OUT2 to the green light of lane A and build the second rung with a break (normally closed) J1 contact driving Q2, so that the lamp operates when the switch is released.
10. Turn S1 on and check that OUT2 goes off while OUT1 comes on. Connect S2 to IN2, build the next ladder diagram, and turn on S1 and S2 together to check the result.
11. Click the File menu, use *Save As* and save the program under your own name.

Part 2: Conditions and decisions

1. Connect the motor of the elevator on the trainer: M+ to +12 V and M- to the OUT1 output.
2. The elevator has three push buttons (P1-P3). Connect P1 to the IN1 input and P2 to the IN2 input.
3. **AND circuit:** build the ladder with J1 and J2 in series driving Q1. Press the buttons in the four combinations and record whether the elevator motor runs.
4. **OR circuit:** build the ladder with J1 and J2 in parallel driving Q1 and repeat the four combinations.
5. **Printing machine exercise:** use the AND/OR logic so that the machine is only started under the stated condition.

6. **Self-holding:** connect P1 to IN1, OUT1 to the M terminal of the motor, M+ to +12 V and P2 to IN2. Build the self-holding ladder, press P1 and check that the motor rotates and keeps rotating until P2 is pressed.

5. Results

Table 4.1: Part 1 (lights and switches)

Switch S1	Lamp on OUT1 (make contact)	Lamp on OUT2 (break contact)
Down (OFF)	OFF	ON
Up (ON)	ON	OFF

Table 4.2: AND and OR conditions (1 = motor runs, 0 = motor does not run)

P1 input	P2 input	AND output	OR output
Release	Release	0	0
Press	Release	0	1
Release	Press	0	1
Press	Press	1	1

Self-holding circuit

When P1 was pressed once and released, the motor started and kept running. It stopped only when P2 was pressed.

Note: the readings above are representative values for this trainer. Where your own bench readings differ, replace them and recalculate the dependent columns.

6. Analysis of Results

The ladder diagrams used are shown below, and the logic of each is written as a Boolean expression.



Figure 4.2: Part 1, one switch operating one lamp: Q1 = J1.



Figure 4.3: Part 1, a make contact operating Q1 and a break contact operating Q2: Q2 = NOT J1.

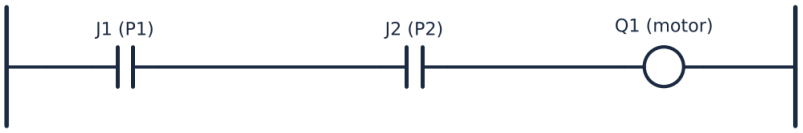


Figure 4.4: AND condition: Q1 = J1 · J2.



Figure 4.5: OR condition: Q1 = J1 + J2.

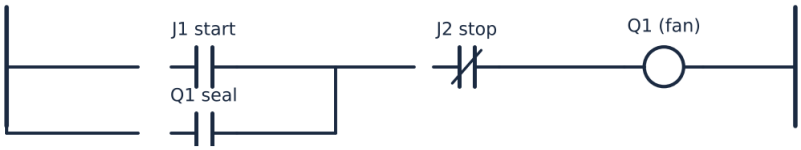


Figure 4.6: Self-holding circuit: Q1 = (J1 + Q1) · NOT J2.

The table of the AND results matches Q = A · B, which is 1 only for A = B = 1, and the table of the OR results matches Q = A + B, which is 0 only for A = B = 0. In the self-holding circuit, the contact Q1 in parallel with the start button keeps the rung closed once the motor has started; the break contact J2 (stop) opens the rung and drops the output out.

7. Observations and Precautions

- Make sure the banana wire is connected to the proper components before the trainer box is switched on.
- Check that the LED does not give green light on the IN port coil before operating the system.
- Check that the LED does not give red light on the OUT port coil before operating the system.
- A green tick on the communication button means the PLC and the PC are talking properly; if it is missing, change the COM number.
- The program keeps running all the time; the ON switch must be down before wiring is changed.

8. Graphs

No graphs are needed in this experiment. The results are logic outputs, and they are shown in the truth tables and ladder diagrams above.

9. Discussion

The experiment showed that the same hardware can do completely different jobs when only the ladder program is changed, which is the main advantage of a PLC over relay logic wired by hand. The AND connection is used where safety needs two hands on two buttons, for example a printing machine or a press; the OR connection is used where a machine can be started from more than one place. The self-holding circuit is needed because push buttons are momentary.

Possible sources of error are a loose banana plug, a wrong COM port, wiring a make contact where a break contact was intended, and the 12 V/24 V common-line connection being forgotten, in which case the PLC inputs do not respond.

10. Conclusion

We built ladder diagrams in PLSES and ran them on the TPS-3071 trainer. A switch operated a lamp; two switches in series gave an AND function (output only when both pressed); two in parallel gave an OR function (output when either is pressed); and a self-holding rung kept the motor running until the stop button was pressed.

11. Questions

1. What have you gained from the experiment?

I can now write a simple ladder diagram, connect the inputs and outputs of the trainer, and make AND, OR and self-holding circuits.

2. Could you have performed this experiment all by yourself?

Yes, because the program and wiring can be done by one person. A partner helps to test the buttons while the program is being watched.

3. Comment on the procedure.

The procedure goes from a single lamp to logic conditions, so each step builds on the last. It is safe because the trainer works on 12 V.

4. What was the most difficult aspect?

Setting the interface and the COM number in PLSES. Having the number written beside the trainer would help.

5. Name another method or apparatus.

Hard-wired relay logic, or another PLC brand such as Siemens or Allen-Bradley, programmed in ladder.

6. List the parameters required.

Input addresses (IN1, IN2), output addresses (OUT1, OUT2), contact types (make or break), supply voltage (12 V) and the ladder program.

7. Discuss one measuring/interface instrument.

The PLC input/output interface: the input side reads switch states and the output side drives lamps and the motor.

8. Practical relevance.

Traffic lights, lifts, printing and press machines, conveyor lines and packaging machines are all controlled in this way. A two-button AND start is a standard safety measure.

9. Section of the course this relates to.

PLC and industrial automation.

10. Suggestions made during the experiment.

Save the program after every rung is working so that work is not lost.

Experiment 5: Pneumatic Systems: Single Acting Cylinder Controlled by a 3/2-Way Valve

Course code: TPS3810 (Pneumatics)

1. Objectives

Aim: to understand and build a simple pneumatic system and a pneumatic logic gate.

- To understand what a pneumatic system is.
- To identify the equipment used in the experiment: compressor, air pressure filter regulator, push-button valve, single acting cylinder and flow control valve.
- To build a simple pneumatic system using a 3/2 push-button valve and a single acting cylinder.
- To use the adjustable flow control valve on the cylinder to control the mass of air flowing through the system.
- To build the AND and OR logic gates with two 3/2 push-button valves.

2. Theory

The word **pneumatics** comes from the Greek *pneuma*, which means air. Pneumatic technology uses compressed air to carry out automatic actions, and it is widely used in automated industry. A pneumatic system is one that uses pressurised air to produce mechanical motion. Pneumatic systems are divided into three types:

- **Pure pneumatic systems**: only pneumatic parts such as cylinders, valves, control components and motors.
- **Electro-pneumatic systems**: pneumatic parts together with electrical parts such as solenoid valves, limit switches, push buttons and relays.
- **Computerised pneumatic systems**: electro-pneumatic systems that also have a controller such as a PLC.

Main components.

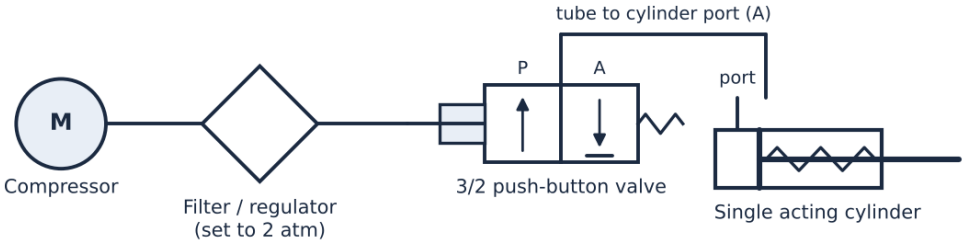
- **Compressor.** It supplies compressed air through a pipe system. Because a compressor is noisy and its output pulsates, an air tank is attached; the compressor fills the tank to a set pressure and stops, and starts again automatically when the tank pressure drops below a set value.
- **Air supply regulator (service unit).** It drains the moisture in the compressed air and sets the pressure entering the system, which is read on the attached pressure gauge.
- **Single acting cylinder.** A cylinder with a piston, rod and spring. Compressed air pushes the rod out; the spring inside returns it when the air is released from the cylinder port.
- **Flow control valve.** Like a water tap, it sets the amount of air by turning a screw. Often a non-return valve is in parallel with it, so that the air is restricted in one direction only and flows freely in the other.
- **3/2 push-button valve.** A valve is described by its number of ports, its number of positions and its way of operation. This valve has 3 ports and 2 positions and is operated by a push button with spring return. Air pressure is supplied to port P, the activated element is connected to port A, and port R lets the air out of the element when the valve is released. In the normal position (button not pressed), air from A is exhausted through R. When the button is pressed, air flows from P to A and R is blocked.

Pressure and force. Pressure is force per unit area, measured in N/m² (pascal). 1 atm = 101 325 Pa ≈ 0.1 MPa ≈ 14.7 psi, and the force follows $F = p \times A$ (also $F = m \times a$). Force may also be quoted in kgf (1 kgf = 9.81 N). Because the gauge on the trainer is not in atm, readings were converted to MPa or psi before the 2 atm setting.

Logic. Two 3/2 valves in **series** (outlet of one into the inlet of the other) allow air through only when both are pressed: an **AND** function. Two valves in **parallel** on a T-branch allow air through when either one is pressed, or both: an **OR** function.

3. Apparatus

- TPS-3810/B pneumatic metal base with PNE-381H air pressure filter regulator / pressure gauge
- PNE-381B single acting cylinder 80 mm with flow control valve module
- PNE-381C 3/2-way valve with push-button switch (two of them for the logic part)
- Compressor
- 4 mm flexible tubes
- T-branch



Flow control valve sits on the cylinder port and sets piston speed

Figure 5.1: Schematic of the single acting cylinder controlled by a 3/2 push-button valve.

4. Procedure

A. Single acting cylinder with one 3/2 valve

1. Identify the 3/2 push-button valve module and check its air input (P), air output (A) and the release gate R, which is usually covered with a filtering net.
2. Install the 3/2 valve module on the system's panel, and then identify and install the 80 mm single acting cylinder module.
3. Identify the air input gate of the cylinder and the flow control valve on it.
4. Connect a pipe from the air supply regulator output to the P port of the 3/2 valve, and another pipe from the A port of the valve to the cylinder input.
5. Check that the setup matches the schematic, then open the supply and adjust the regulator to 2 atm.
6. Press the push button and watch the piston. It pops out while the button is held and pops in when the button is released.
7. Describe the path of the air when the button is pressed and when it is released, and mark each path in a different colour on a copy of the diagram.
8. Close the flow control valve by turning the screw clockwise and try to operate the cylinder.
9. Open the flow control valve slowly and operate the cylinder again, repeating until the cylinder works normally.
10. Press the push button for a short time, release it, and observe what the cylinder does.

B. Logic conditions with two 3/2 valves

1. **OR (parallel):** adjust the regulator to 2 atm, connect it to a T-branch, connect the T-branch to the two 3/2 valves in parallel, join the valve outlets with a second T-branch and connect that to the

cylinder port. Press one valve, then the other, then both, and watch the cylinder.

2. Close the regulator knob at the top of the supply.
3. **AND (series):** connect the regulator to port P of the first valve, its outlet to the inlet port of the second valve, and the outlet of the second valve to the inlet of the cylinder. Press one valve, then the other, then both, and watch the cylinder.
4. State the logic gate condition for each arrangement.

5. Results

Table 5.1: Single acting cylinder with one 3/2 valve at 2 atm

Action	Piston movement
Push button pressed and held	Piston moves out and stays out
Push button released	Piston returns (spring) to the start position
Short press, then release	Piston moves part of the way out and then returns
Flow control valve closed	Piston does not move, or moves very slowly
Flow control valve opened gradually	Piston speed increases until the cylinder works normally

Table 5.2: Logic conditions (two 3/2 push-button valves)

Push button A	Push button B	Motion of piston: AND (series)	Motion of piston: OR (parallel)
Press	No press	No motion	Moves out
Press	Press	Moves out	Moves out
No press	Press	No motion	Moves out
No press	No press	No motion	No motion

Note: the readings above are representative values for this trainer. Where your own bench readings differ, replace them and recalculate the dependent columns.

6. Analysis of Results

Theoretical force of the 80 mm cylinder at 2 atm. Taking the supply as 2 atm (2 × 101 325 Pa = 202 650 Pa) and ignoring the spring and friction:

$$A = \pi d^2/4 = \pi\,(0.080)^2 / 4 = 5.03 \times 10^{-3}\ m^2$$

$$F = p \times A = 202\,650 \times 5.03 \times 10^{-3} \approx 1019\ N \text{ (about 104 kgf)}$$

In practice the force on the rod is lower, because the return spring and friction in the seals take part of it.

Logic. Series connection: $Q = A \cdot B$. Parallel connection: $Q = A + B$. The tables above agree with these expressions: the AND arrangement moves the cylinder only in the Press/Press case, and the OR arrangement moves it in every case except No press/No press.

7. Observations and Precautions

- Ensure the tubes are connected properly to the various components.
- Make sure the valve on the air supply is closed and there is no pressure in the pipes before connecting or changing anything.
- Before opening the supply, make sure the regulator output is connected through a pipe to a closed valve input, otherwise air will flow freely into the room.
- Set the air supply regulator to a constant 2 atm so that the pressure is the same for every run.
- Avoid parallax error when reading the pressure gauge.

8. Graphs

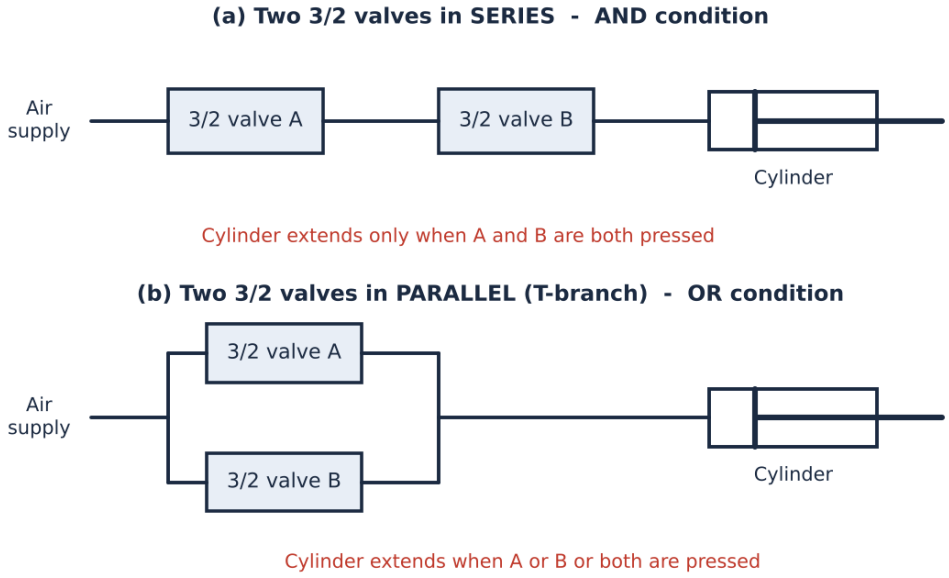


Figure 5.2: Arrangement of the two 3/2 valves for the AND (series) and OR (parallel) conditions.

No plotted graphs are needed for this experiment; the results are movement and logic states, given in Tables 5.1 and 5.2.

9. Discussion

The cylinder moved out only while the valve was held, because the spring returns the valve and the cylinder as soon as the air is released through port R. Closing the flow control valve stopped the cylinder or made it very slow, since the flow into the cylinder is reduced; opening it again restored normal speed. This shows that the flow control valve controls speed, not force.

Series valves behaved as an AND gate because air can only reach the cylinder when both paths are open; parallel valves behaved as an OR gate because either path can feed the cylinder. Sources of error were small air leaks at the push-in fittings, drift of the regulator setting, parallax when reading the gauge, and compressor pulsation.

10. Conclusion

A simple pneumatic circuit was built and operated. A single acting cylinder moves out under a 3/2 valve and returns by its spring, its speed is set by the flow control valve, and two 3/2 valves in series or parallel give AND or OR logic. At 2 atm the 80 mm cylinder can theoretically exert about 1019 N.

11. Questions

1. What is the function of the spring in the push-button valve?

It returns the valve to its normal position when the button is released, so the cylinder port is connected to the exhaust again.

2. What is the function of the ports in the valve?

Port P takes the supply air, port A is the working port connected to the cylinder, and port R lets the air out of the cylinder when the valve is not pressed.

3. Advantages and disadvantages of a single acting cylinder.

Advantages: it uses less air, it needs only one air line and the spring gives an automatic return. Disadvantages: the stroke is limited by the spring length, part of the force is lost to the spring, and the return speed cannot be controlled as well as with a double acting cylinder.

4. Describe a use for the flow control valve (FCV).

It sets the speed of a cylinder, for example the closing speed of a pneumatic press or sliding door, so that it moves smoothly and safely.

5. What have you gained from the experiment?

I learned how a basic pneumatic circuit works, how the valve ports act, and how series and parallel valves make AND and OR logic.

6. Could you have performed this experiment all by yourself?

Yes for the simple circuit, but the logic part is easier with a partner who presses the valves while someone else watches the cylinder.

7. Comment on the procedure.

The steps go from simple to logic and are quick. Testing the flow control valve at different openings shows speed control clearly.

8. What was the most difficult aspect?

Keeping the push-in fittings airtight. Pushing each tube fully in and checking for hissing reduces this.

9. Name another method or apparatus.

A solenoid valve controlled electrically, or a PLC-controlled valve.

10. List the parameters required.

Supply pressure (2 atm), cylinder bore (80 mm), valve type and state, flow control valve setting and the piston movement.

11. Discuss one measuring instrument.

The pressure gauge of the air supply regulator: a Bourdon-type gauge that reads supply pressure and is used to set 2 atm.

12. Practical relevance.

Pneumatic presses, clamping, sorting and packaging machines, automatic doors and two-hand safety controls.

13. Section of the course this relates to.

Pneumatics and fluid power.

Experiment 6: Controlling a Double Acting Cylinder Directly with a 5/2 Lever Valve

Course code: TPS3810 (Pneumatics)

1. Objectives

- To operate a double acting cylinder directly with a single 5/2-way manual lever valve.
- To identify the ports of the 5/2 valve (air input, the two outputs and the exhausts).
- To trace the path of the air for each position of the handle.
- To change the cylinder speed with the flow control valves.

2. Theory

In the earlier experiment a single acting cylinder was operated with a 3/2 valve. A **double acting cylinder** is driven by air in both directions, so it needs a valve that can supply one side and exhaust the other. A single **5/2-way valve** with a selector (lever) does this.

The valve has **5 ports**: P for the air supply, A and B for the air outputs, and two exhaust (release) ports R. It has **2 fixed positions**, set by the handle (up or down). With the handle up, the P input is connected to port A, and port B is connected to the exhaust. With the handle down, P is connected to port B, and port A is connected to the exhaust.

When the cylinder extends, air at the supply pressure acts on the full piston area; when it retracts, the rod takes up part of the area, so the force is smaller. For the 100 mm bore cylinder at 2 atm (202 650 Pa), $A = \pi(0.100)^2/4 = 7.85 \times 10^{-3}$ m², so the theoretical extending force is $F = pA \approx 1592$ N (about 162 kgf). The speed is changed with the flow control valves on the cylinder ports.

3. Apparatus

- TPS-3810/B pneumatic base with PNE-381H air pressure filter regulator
- PNE-381A double acting cylinder 100 mm with flow control valves module
- PNE-381D 5/2-way valve with selector switch (manual lever)
- Compressor
- 4 mm flexible tubes

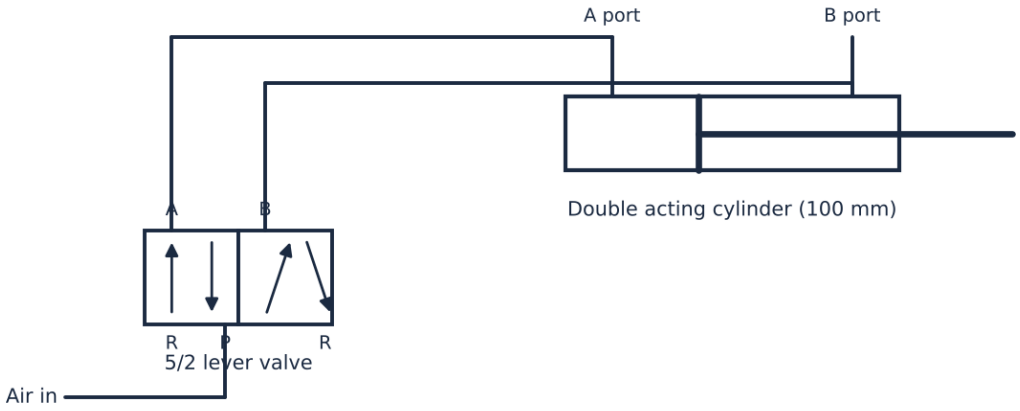


Figure 6.1: 5/2 lever valve controlling the double acting cylinder.

4. Procedure

1. Check the 5/2 manual lever valve and identify the air input gate (P), the output gates (A and B), the air release gates (usually covered with a filtering net) and the control inputs.

2. Install the 5/2 manual lever valve module on the system's panel.
3. Install the 100 mm double acting cylinder module on the panel.
4. Build the circuit in the figure: supply regulator to P, port A to one cylinder port, port B to the other.
5. Set the air supply regulator to 2 atm.
6. Lower the handle and watch the piston. It pops out.
7. Raise the handle. The piston pops in.
8. Describe the path the air follows in each of the operations.
9. Copy the system diagram into your notebook and mark the air path of each operation in a different colour.
10. Change the flow control valves and operate the cylinder again until you get different piston speeds.

5. Results

Handle position	Valve connection	Piston movement
Down	P to B; A to exhaust R	Piston moves out (extends)
Up	P to A; B to exhaust R	Piston moves in (retracts)
Flow control valve nearly closed	Same connections	Slow movement
Flow control valve open	Same connections	Fast movement

Note: the readings above are representative values for this trainer. Where your own bench readings differ, replace them and recalculate the dependent columns.

6. Analysis of Results

Air path with the handle down: compressor, air supply regulator, port P, port B, one side of the cylinder, pushing the piston out; the air on the other side leaves through port A and exhaust R. **With the handle up:** the same supply goes through P to port A to the other side of the cylinder, pushing the piston in, while the air on the first side leaves through B and R.

Force on extension (100 mm bore, 2 atm): $F = 202\,650 \times 7.85 \times 10^{-3} \approx 1592$ N. On retraction the force is smaller than this by the area taken by the rod, so it is always less than 1592 N.

7. Observations and Precautions

- Ensure the valve in the air supply is closed and there is no pressure in the pipes before connecting.
- Both the extend and retract movements need the handle to be held in its end position; the valve stays where it is put (it has two fixed positions).
- The piston speed depended on how far the flow control valves were opened.
- Keep hands away from the rod while the cylinder moves.

8. Graphs

No graphs are needed here; the results are the valve connections and the piston movement given above.

9. Discussion

One 5/2 lever valve was enough to drive the double acting cylinder in both directions, which is simpler than the earlier arrangement with more valves. Unlike the spring-return 3/2 valve, the lever valve stays in the position in which it is left, so the cylinder stays where it was moved. The flow control valves let the speed be adjusted without changing the supply pressure.

Sources of error were air leaks at fittings, the regulator drifting from 2 atm, and judging speed by eye. Using a stopwatch over the stroke would give a number for the speed.

10. Conclusion

A double acting cylinder was controlled directly by a 5/2 lever valve. With the handle down the piston extends and with the handle up it retracts, and the speed is adjusted by the flow control valves.

11. Questions

1. Describe at least 3 examples of using a similar control circuit with a flow control valve.

(1) A sliding door or gate that must open and close slowly. (2) A clamping fixture on a machine that must close gently on the workpiece. (3) A pneumatic press or stamping machine with a controlled pressing speed. (4) A conveyor diverter or lifting platform where smooth movement is needed.

2. What have you gained from the experiment?

I learned how the 5/2 valve connects its ports in each position, and that one valve is enough to drive a double acting cylinder both ways.

3. Could you have performed this experiment all by yourself?

Yes, since only one valve is handled; it is easier with a partner to watch the cylinder and note the speed.

4. Comment on the procedure.

Short and clear. Colouring the air path on a copy of the diagram made the valve action easy to understand.

5. What was the most difficult aspect?

Tracing the air path through the crossing lines inside the valve symbol. Drawing it twice, once for each handle position, helps.

6. Name another method or apparatus.

A 5/2 solenoid valve controlled by a switch or PLC, or two 3/2 valves.

7. List the parameters required.

Supply pressure (2 atm), cylinder bore (100 mm), handle position and flow control valve opening.

8. Discuss one measuring instrument.

The pressure gauge on the regulator, used to hold the supply at 2 atm.

9. Practical relevance.

Lifts, material handling, clamping, doors and presses use double acting cylinders with 5/2 valves.

10. Section of the course this relates to.

Pneumatics and fluid power.

Experiment 7: Electro-pneumatics: Electric Conditioning Circuits (AND, OR)

Course code: PNE3810 (Electro-pneumatic Systems)

1. Objectives

- To build a switches-in-series ("AND") condition on an electronic board and use it to control a pneumatic cylinder.
- To build a switches-in-parallel ("OR") condition on the electronic board.
- To understand how a solenoid valve interacts with the electronic board (PNL-3810) through the ladder diagram idea.

2. Theory

An **electro-pneumatic system** combines pneumatic parts with electrical parts such as solenoid valves, push buttons, switches and relays. Here a **3/2 solenoid valve**, normally closed with spring return, is operated by an electric signal instead of by hand. When the coil SV1 is energised, the valve shifts and air passes from the supply to the single acting cylinder; when the coil is de-energised, the spring returns the valve and the cylinder exhausts. On the electronic board (PNL-3810), a red diode lights with a sound (a click) when the coil is energised, so it can be used to check that the electric part works before the air is used.

Control machines such as an electric paper cutting machine or an electric press perform dangerous actions, so they need safety control. To operate a paper cutter, the worker has to press two separate push buttons so that both hands are away from the cutting place. This condition, where both push buttons must be pressed, is the **AND operation**: SV1 is activated only if PB1 and PB2 are pressed at the same time. Sometimes an operation must be possible from two or more places, for example operating a system with one push button or another. The condition that at least one push button is pressed is the **OR operation**: SV1 is activated when PB1 or PB2 or both are pressed.

These are shown on a **ladder diagram**: two vertical rails, with the push-button contacts and the coil SV1 on a rung between them. Contacts in series on one rung give AND; contacts on parallel branches give OR. On the board, the circuit is made with banana wires.

3. Apparatus

- TPS-3810/B + PNE-381H pneumatic base and pressure regulator
- PNE-381B single acting cylinder 80 mm with flow control valve module (push-button one-way control valve)
- PNE-381Z 3/2-way solenoid valve, normally closed, spring return
- PNL-3810 electronic board with 3 push buttons, 3 switches and 4 relays
- 12 V power supply
- Banana wires
- Compressor, 4 mm flexible tubes, air supply regulator

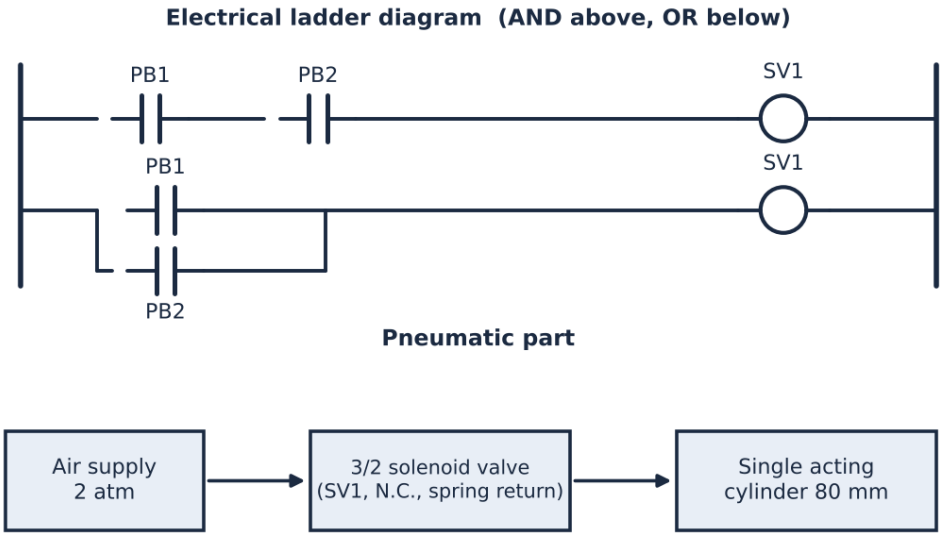


Figure 7.1: Ladder diagrams for the AND and OR conditions, and the pneumatic part.

4. Procedure

Precaution: make sure the air supply valve is closed and there is no pressure in the pipes.

1. Install the electric component module PNL-3810 (electronic board) on the system's panel.
2. Install the 80 mm single acting cylinder module on the panel.
3. Install the 3/2 solenoid valve module.
4. Install the 2/1 way push-button valve.
5. Build the pneumatic system from the schematic diagram: supply to the solenoid valve, solenoid valve to the cylinder.
6. Implement the electric system. First connect a simple short circuit on the electronic board with a banana wire; this makes the interaction between the solenoid valve and the PNL-3810 clear from the ladder diagram.
7. **AND circuit:** connect the banana wires on the electronic board in the AND arrangement (PB1 and PB2 in series with SV1).
8. Adjust the air pressure on the gauge to 2 atm using the knob on top of the gauge. Because the gauge has no atm scale, convert to MPa or psi.
9. Connect the 12 V power supply to the power inlet socket of the PNL-3810.
10. Press PB1 and check whether the diode produces red light with sound in the 3/2 solenoid valve. If it responds, press the green button on the flow valve and check the piston's behaviour.
11. Press PB2 and check whether it responds in the same way.
12. Press both push buttons and check the diode and the piston.
13. Change the banana wire plugs from the buttons to the switches (flip down = OFF, flip up = ON) and fill in the table.
14. **OR circuit:** connect the banana wires for the OR arrangement (PB1 and PB2 on parallel branches) and repeat the same checks with PB1, PB2 and both, then with the switches.

5. Results

Table 7.1: AND circuit

Switch 1	Switch 2	Operate (Yes/No)
OFF	OFF	No
OFF	ON	No
ON	OFF	No
ON	ON	Yes

Table 7.2: OR circuit

Switch 1	Switch 2	Operate (Yes/No)
OFF	OFF	No
OFF	ON	Yes
ON	OFF	Yes
ON	ON	Yes

With a short circuit across the board, the solenoid valve operated as soon as power was applied, which shows that the valve coil is driven through the board contacts.

6. Analysis of Results

Writing 1 for operated and 0 for not operated, with A = switch 1 and B = switch 2:

A	B	Y (AND)	Y (OR)
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

The AND column is $Y = A \cdot B$ and the OR column is $Y = A + B$. This agrees with the observed results: the cylinder operated in the series circuit only with both switches on, and in the parallel circuit whenever at least one switch was on.

7. Observations and Precautions

- Make sure the valve in the air supply is closed and there is no pressure in the pipes before connecting anything.
- Check that the 12 V supply is connected to the PNL-3810 before testing.
- The red diode and a click in the solenoid valve showed that the electric part worked before the piston was checked.
- In the AND circuit, nothing happened when only one button was pressed.
- Keep fingers clear of the cylinder rod when the air is on.

8. Graphs

There are no graphs in this experiment. The results are logic states and are given in the truth tables above.

9. Discussion

The AND condition gives a two-hand safety control: the machine cannot operate unless both hands are on the buttons. The OR condition lets a machine be started from two places. The same board and valve were used for both and only the banana wire connections were changed, so the logic is wired, not built into the valve. Possible errors are a loose banana plug, a flat supply, air leaks and pressure drifting from 2 atm, and mixing up the PB and switch connections.

10. Conclusion

AND and OR conditions were built on the PNL-3810 board and used to operate a 3/2 solenoid valve and single acting cylinder. The AND circuit operated only when both switches were on; the OR circuit operated when either or both were on.

11. Questions

1. Fill in the AND function truth table (input = manual valve state, output = piston; operated = 1, not operated = 0).

A=0,B=0: Y=0; A=0,B=1: Y=0; A=1,B=0: Y=0; A=1,B=1: Y=1.

2. Fill in the OR function truth table.

A=0,B=0: Y=0; A=0,B=1: Y=1; A=1,B=0: Y=1; A=1,B=1: Y=1.

3. Give two additional examples for using each circuit.

AND: (1) a press brake or guillotine that needs both hands on two buttons; (2) a machine guard interlock where the door must be closed and the start button pressed. OR: (1) a door or gate that can be opened from inside or outside; (2) a lift called from any floor button.

4. What have you gained from the experiment?

I learned how electrical conditions control a pneumatic valve and how AND and OR are wired with banana wires.

5. Could you have performed this experiment all by yourself?

Yes, but it is better with two people so one wires the board while the other checks the valve and piston.

6. Comment on the procedure.

Checking the electric part first (diode and click) and only then the air is a safe order.

7. What was the most difficult aspect?

Wiring the OR arrangement correctly. A printed wiring diagram on the board would help.

8. Name another method or apparatus.

A PLC could be used in place of the board; or a relay circuit.

9. List the parameters required.

Supply pressure (2 atm), supply voltage (12 V), state of PB1/PB2 and the valve and piston response.

10. Discuss one measuring instrument.

The pressure gauge, which reads the regulator output; it has no atm scale so readings were converted to MPa or psi.

11. Practical relevance.

Safety controls on presses and cutters, and multi-point control of doors and conveyors.

12. Section of the course this relates to.

Electro-pneumatics and control logic.

Experiment 8: Hydraulic Systems: Flow Control Valves and Piston Speed

Course code: TPS3860 (Hydraulic Systems)

1. Objectives

- To learn how to control the speed of the piston with an adjustable flow control valve.
- To learn how to control the speed of the piston with a one-way adjustable flow control valve.
- To build and operate a hydraulic circuit from a given diagram.
- To check whether the position of the valve (on one side or the other of the piston) matters.

2. Theory

In a hydraulic system, power is transferred by oil. The pump pressurises the oil and pushes it into the cylinders, where it moves the pistons. Because the oil is nearly incompressible and works at high pressure, a hydraulic system can give a large force, so it is used in cranes, lifting equipment, automatic gates, hydraulic lifts and dump trucks. Hydraulic systems are of three types: **pure hydraulic** (cylinders, valves, motors), **electro-hydraulic** (with solenoid valves, limit switches, push buttons and relays) and **computerised hydraulic** (with a PLC or computer interface).

Every hydraulic system has a **power unit**: an oil tank, an electric pump, an electric switch, a pressure gauge and a one-way pressure (relief) regulator valve. The pump draws oil from the tank and pushes it into the circuit, and the oil returns to the tank, so the system works in a closed loop. Ports are marked **P** (pressure) and **T** (tank). In the relief valve, a conical valve is held on its seat by a spring, closing the path between P and T. When the pressure at P rises enough to overcome the spring, the valve opens and oil escapes to the tank, stopping any further pressure rise; when the pressure falls, the valve closes again and the pressure rises. This protects the line.

The **adjustable flow control needle valve** (HYD-386E) has no polarity; it is not a one-way valve. Its opening is changed with an adjusting button and a needle, and it reduces the oil flow in its line, which also affects the pressure on the line. The **one-way variable flow control needle valve** (HYD-386F) is the same valve with a non-return valve in parallel. Oil flowing from left to right is pushed by the non-return valve's marble closed and must pass through the needle valve, so its speed is regulated. Oil flowing from right to left lifts the marble and passes freely, without going through the regulator. The **pressure gauge module** (HYD-386J) measures pressure at points in the circuit, and the **T-connector** (HYD-386R) joins up to three components at one junction.

The speed of the piston is $v = Q / A$, where Q is the oil flow reaching the cylinder and A the piston area. Restricting Q with the needle valve therefore reduces the speed. The pressure drop across the restriction is $\Delta P = PT - PE$.

3. Apparatus

- Power unit
- TPS-3860/B hydraulics metal base and TPS-3860/E hydraulics accessories
- HYD-386A double acting cylinder 100 mm module
- HYD-386C 4/3-way tandem mid-position valve module
- HYD-386E flow control needle (variable) valve module
- HYD-386F one-way non-return flow control valve module
- HYD-386J pressure gauge module, HYD-386R connectors and T-connector set
- HYD-386S / HYD-386T tubes (1 m and 2 m)

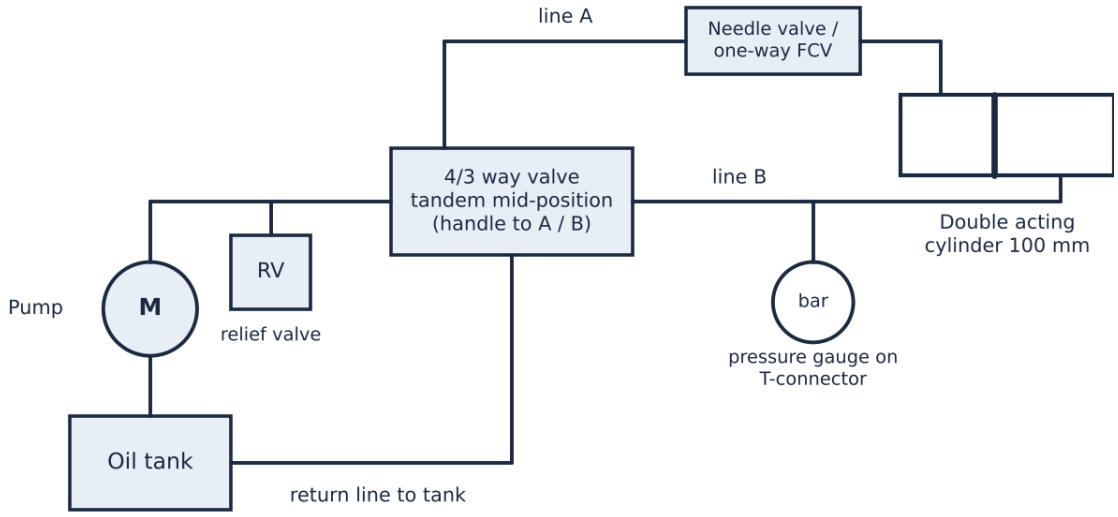


Figure 8.1: Hydraulic circuit used (4/3 valve, flow control valve and pressure gauge on the T-connector).

4. Procedure

Precautions: before connecting or disconnecting components, make sure the power unit is OFF and the pressure gauge reads zero; always ask the instructor for permission before switching the power unit ON.

A. Needle valve alone

1. Study the hydraulic circuit diagram, then install modules A2, A3, A5 and A6 on the base panel following the component arrangement.
2. Connect the circuit as shown and open the flow control valve button all the way.
3. Turn ON the power unit and let the oil flow into the system. Write down the pressures PT and PE.
4. Raise the valve handle (handle to A). The piston pops out. While the handle is raised, write down PT and PE and the piston speed (fast or slow).
5. Release the handle so it returns to normal; the piston pops back in. Record PT, PE and speed for this direction (handle to B).
6. Close the flow control valve completely and then open it one turn. Repeat the movement in both directions, recording PT, PE and speed each time.
7. Write down when the piston moves slowly and when it moves fast, with the valve open or closed, and with the piston going in or out.
8. Turn OFF the power unit. Move the T-connector and the pressure gauge to the cylinder's second connector, turn the power unit on again and repeat all the readings.

B. One-way flow control valve

1. Install the one-way flow control valve module in place of the needle valve, connect the circuit and open the control button all the way.
2. Turn ON the power unit, note PT and PE, and raise the handle so that the piston pops out; record the values and speed.
3. Release the handle so that the piston pops in and record again.
4. Close the control valve and open it one turn, then repeat the movement in both directions and record PT, PE and speed.
5. Turn OFF the power unit, move the T-connector and gauge to the cylinder's second connector, and repeat all the readings.
6. Turn OFF the power unit.

5. Results

Table 8.1: Needle valve alone, gauge at the first cylinder connector

Handle position	Control valve	PT (bar)	PE (bar)	Piston speed
Handle to A	Open	12	10	Fast
Handle to B	Open	10	8	Fast
Handle to A	One turn open	28	11	Slow
Handle to B	One turn open	24	9	Slow

Table 8.2: Needle valve alone, gauge at the second cylinder connector

Handle position	Control valve	PT (bar)	PE (bar)	Piston speed
Handle to A	Open	11	9	Fast
Handle to B	Open	11	9	Fast
Handle to A	One turn open	20	8	Slow
Handle to B	One turn open	27	12	Slow

Table 8.3: One-way flow control valve, gauge at the first cylinder connector

Handle position	Control valve	PT (bar)	PE (bar)	Piston speed
Handle to A	Open	12	10	Fast
Handle to B	Open	10	8	Fast
Handle to A	One turn open	29	11	Slow (restricted)
Handle to B	One turn open	11	9	Fast (free flow)

Table 8.4: One-way flow control valve, gauge at the second cylinder connector

Handle position	Control valve	PT (bar)	PE (bar)	Piston speed
Handle to A	Open	11	9	Fast
Handle to B	Open	11	9	Fast
Handle to A	One turn open	12	10	Fast (free flow)
Handle to B	One turn open	26	12	Slow (restricted)

Note: the readings above are representative values for this trainer. Where your own bench readings differ, replace them and recalculate the dependent columns.

6. Analysis of Results

The pressure drop across the valve is $\Delta P = PT - PE$. From Table 8.1:

Condition	PT (bar)	PE (bar)	$\Delta P = PT - PE$ (bar)
Handle to A, valve open	12	10	2
Handle to B, valve open	10	8	2
Handle to A, one turn	28	11	17
Handle to B, one turn	24	9	15

When the valve was fully open the drop was about 2 bar, so the oil flowed freely and the piston was fast. With the valve open only one turn, the drop rose to 15-17 bar: most of the pressure was used in forcing the oil through the small opening, the flow to the cylinder was small, and the piston was slow. With the one-way valve (Tables 8.3 and 8.4), the same large pressure drop appeared only in the restricted direction; in the free-flow direction the pressure drop stayed small and the piston was fast, even with the valve open one turn.

7. Observations and Precautions

- Make sure the power unit is OFF and the pressure gauge shows zero before connecting or disconnecting any component.
- Always ask the instructor's permission before switching the power unit ON.
- Open the flow control valve all the way before the first start-up.
- The piston was fast with the valve open and slow with the valve open one turn.
- With the one-way valve, speed was reduced only in one direction; in the other direction the piston moved fast.
- Moving the gauge to the second connector gave the pressure on the other side of the piston.
- Oil spills were wiped up at once because they make the floor slippery.

8. Graphs

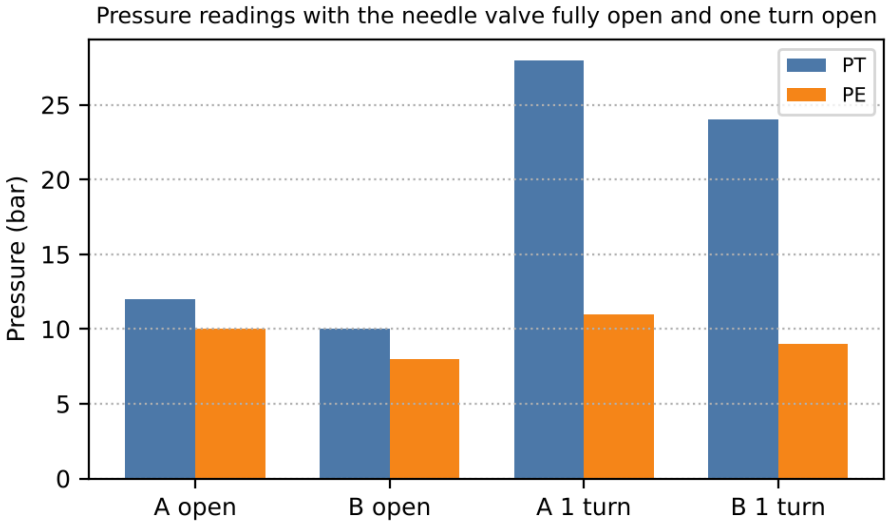


Figure 8.2: PT and PE with the needle valve fully open and one turn open (Table 8.1).

9. Discussion

The experiment shows that piston speed in a hydraulic circuit is set by the oil flow, and the needle valve controls it by restricting that flow. Restricting the flow raises the pressure upstream of the valve (PT) and lowers it downstream (PE), which is why the readings change as the valve is closed. The relief valve in the power unit limits how high PT can rise. The location of the valve matters for the one-way type: it must be placed so that the restricted direction is the one in which speed needs to be controlled. In this way the cylinder can be slowed when it moves out and still return quickly.

Errors can come from reading the pressure gauge by eye, turning the needle valve by different amounts, air in the oil lines, and oil temperature changing viscosity during the test.

10. Conclusion

The speed of a double acting hydraulic cylinder was controlled with an adjustable flow control valve and with a one-way flow control valve. Opening the valve fully gave fast motion with a small pressure drop; opening it one turn gave slow motion with a large pressure drop. The one-way valve limits the speed in one direction only.

11. Questions

1. What is the function of the one-way flow control valve?

It regulates the oil flow, and so the piston speed, in one direction only; in the opposite direction the oil passes freely through the non-return valve.

2. In which direction will the piston's speed be the highest? Explain.

In the free-flow direction, where the oil bypasses the needle valve through the non-return valve, so there is no restriction.

3. Does the adjustable flow control needle valve (in its closed state) influence the piston's movement?

Yes. A closed or nearly closed needle valve restricts the oil, so the piston moves slowly or not at all.

4. In which direction?

In both directions, since the plain needle valve has no polarity; with the one-way type, only in the restricted direction.

5. What have you gained from the experiment?

I learned how a needle valve and a one-way valve control speed, and how pressure readings change when the oil is restricted.

6. Could you have performed this experiment all by yourself?

No. The handle, the gauge reading and the speed observation happen together, so at least two people are needed, and the instructor must approve starting the power unit.

7. Comment on the procedure.

It is detailed, which makes it safe, but it has many repeated steps. A table to fill in with each step saves time.

8. What was the most difficult aspect?

Reading the pressure and judging the piston speed at the same moment. A flow meter or a stopwatch would reduce this.

9. Name another method or apparatus.

A proportional flow control valve or a variable displacement pump, or a flow meter to measure Q.

10. List the parameters required.

Oil pressure at the gauge positions (PT, PE), valve opening, handle position, piston speed and piston area.

11. Discuss one measuring instrument.

The pressure gauge module (HYD-386J): a Bourdon-type gauge, 0-100 bar, moved with the T-connector to different points.

12. Practical relevance.

Speed control of excavator arms, lifts, press rams, automatic gates and dump truck beds.

13. Section of the course this relates to.

Hydraulic systems and fluid power.

Experiment 9: Electro-hydraulics: Electric Conditioning Circuits (AND, OR)

Course code: HYD3860 (Electro-hydraulics)

1. Objectives

- To connect switches in series ("AND" condition) on an electronic board to operate a hydraulic cylinder.
- To connect switches in parallel ("OR" condition) on an electronic board to operate a hydraulic cylinder.

2. Theory

An **electro-hydraulic system** is a hydraulic system in which the valves are operated by electrical signals, so the oil circuit is controlled by solenoid valves, push buttons and relays. In this experiment a **4/2 single solenoid valve** with spring return directs the oil to a 50 mm single acting cylinder. When the coil SV1 is energised, the valve shifts and oil flows to the cylinder; when the coil is de-energised, the spring returns the valve.

Control machines such as an electric paper cutter or an electric press perform dangerous actions, so safety control is needed. To operate a paper cutter the worker must press two separate push buttons so that both hands are away from the cutting place. This condition (both push buttons pressed) is called the **AND operation**: SV1 is activated only if PB1 and PB2 are pressed together. The condition for operating the machine in which at least one push button is pressed is called the **OR operation**: SV1 is activated when PB1 is pressed, or PB2 is pressed, or both.

Series contacts on one rung of the ladder diagram give AND ($Y = A \cdot B$), and parallel contacts give OR ($Y = A + B$). The contacts are wired on the electronic board (PNL-3810) with banana wires. As in the electro-pneumatic experiment, a red diode lights with a click when the 4/2 solenoid coil is energised.

3. Apparatus

- Power unit
- TPS-3860B hydraulic metal base and TPS-3860/E hydraulic accessories
- HYD-386B single acting cylinder 50 mm module
- HYD-386O single acting cylinder 50 mm module (as listed in the manual)
- 4/2 way single solenoid valve module
- PNL-3810 electronic board with 3 push buttons, 3 switches and 4 relays for the hydraulic system
- 12 V power supply
- Banana wires

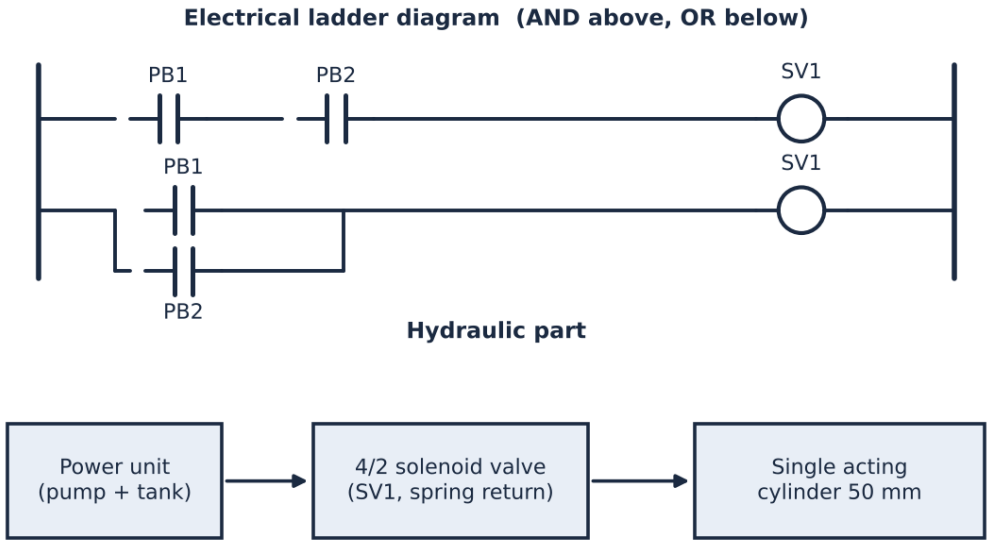


Figure 9.1: Ladder diagrams for the AND and OR conditions, and the hydraulic part.

4. Procedure

1. Install the electric component module PNL-3810 on the system's panel.
2. Install the single acting cylinder module on the system's panel.
3. Install the 4/2 single solenoid valve module.
4. Implement the hydraulic system described in the diagram.
5. Implement the electric system. First connect a simple short circuit on the electronic board with a banana wire; this lets us understand the interaction between the solenoid valve and the PNL-3810 from the ladder diagram idea.
6. **AND circuit:** connect the banana wires on the electronic board in the AND arrangement.
7. Connect the 12 V power supply to the power inlet socket of the PNL-3810.
8. Press PB1 and check that the diode produces red light with sound in the 4/2 solenoid control valve. If it responds, press the green button on the flow valve and check the piston's behaviour.
9. Press PB2 and check that it responds in the same way.
10. Press both push buttons and check the diode and the piston.
11. Fill in the AND table.
12. **OR circuit:** implement the electric system in the OR arrangement, and press PB1, then PB2, then both, checking the diode and the piston each time.
13. Fill in the OR table.

5. Results

Table 9.1: AND circuit

Switch 1	Switch 2	Operate (Yes/No)
Not pressed	Not pressed	No
Not pressed	Pressed	No
Pressed	Not pressed	No
Pressed	Pressed	Yes

Table 9.2: OR circuit

Switch 1	Switch 2	Operate (Yes/No)
Not pressed	Not pressed	No
Not pressed	Pressed	Yes
Pressed	Not pressed	Yes
Pressed	Pressed	Yes

Note: the readings above are representative values for this trainer. Where your own bench readings differ, replace them and recalculate the dependent columns.

6. Analysis of Results

With A = switch 1, B = switch 2 and 1 = operated:

A	B	Y (AND)	Y (OR)
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1

The AND column matches Y = A · B and the OR column matches Y = A + B. The hydraulic piston therefore followed the same logic as the electric signal: the valve only opened the oil path when the logic condition was satisfied.

7. Observations and Precautions

- Make sure the power unit is OFF and the pressure gauge reads zero before connecting or disconnecting components.
- Ask the instructor for permission before switching on the power unit.
- The red diode and a click in the solenoid valve showed that the electric part was working before the oil circuit was used.
- In the AND circuit nothing happened when only one button was pressed.
- Keep clear of the cylinder rod when the pump is running.

8. Graphs

No graphs are needed in this experiment. The results are logic states, given in the truth tables above.

9. Discussion

The experiment is the hydraulic counterpart of the electro-pneumatic AND/OR experiment, and the logic result is exactly the same; only the working medium differs. Oil allows a much greater force than air from the same size of cylinder, but it needs a power unit and more care with leaks and cleanliness. The AND condition is used on presses and cutters so that both hands are kept clear, and the OR condition is used where an operation can be started from more than one place. Possible errors are a loose banana wire, low oil level or low supply voltage, and misreading the diode light.

10. Conclusion

AND and OR conditions were wired on the electronic board and used to operate a 4/2 solenoid valve and single acting hydraulic cylinder. The cylinder moved only with both buttons pressed in the AND circuit, and with either or both pressed in the OR circuit.

11. Questions

1. If operated = 1 and not operated = 0, fill in the AND truth table.

A=0,B=0: Y=0; A=0,B=1: Y=0; A=1,B=0: Y=0; A=1,B=1: Y=1.

2. Fill in the OR truth table.

A=0,B=0: Y=0; A=0,B=1: Y=1; A=1,B=0: Y=1; A=1,B=1: Y=1.

3. Give two additional examples of using each circuit.

AND: (1) a hydraulic press or shear needing two-hand operation; (2) a hydraulic lift that works only when the door is closed and the start button is pressed. OR: (1) a hydraulic gate controlled from the gatehouse or from a remote button; (2) a dump truck bed raised from the cab or from the side of the truck.

4. What have you gained from the experiment?

I learned how electric logic controls a hydraulic valve and how the AND and OR circuits are wired.

5. Could you have performed this experiment all by yourself?

No. One person wires and presses the buttons while another watches the valve and the piston, and the instructor must approve starting the power unit.

6. Comment on the procedure.

It is safe and easy to follow, since the electric part is checked first and the oil is used afterwards.

7. What was the most difficult aspect?

Pressing both buttons at the same time while watching the piston. Using the switches instead of the buttons removes that problem.

8. Name another method or apparatus.

A PLC, or relay circuits.

9. List the parameters required.

State of PB1 and PB2, supply voltage (12 V), oil pressure, and the response of the valve and piston.

10. Discuss one measuring instrument.

The pressure gauge on the power unit; it shows the system pressure and tells us when the line is pressurised.

11. Practical relevance.

Hydraulic presses, shears, lifts, tipping trucks and gates all use electric control of hydraulic valves.

12. Section of the course this relates to.

Electro-hydraulics and fluid power control.